

Iteration 1 by Racheal Resty Akello

$$X = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \quad Y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$$

$$k_t = \begin{pmatrix} m_{1,t} \\ m_{2,t} \end{pmatrix} \quad B_t = \begin{pmatrix} b_{1,t} \\ b_{2,t} \end{pmatrix}$$

$$\hat{y}_t = X k_t + B_t$$

$$E_t = \hat{y}_t - Y$$

$$\Delta k_t = \frac{1}{2} X^T E_t, \quad \nabla B_t = \frac{1}{2} E_t$$

$$\alpha (\text{Learning rate}) = 0.1$$

Iteration 1

$$k_0 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\hat{y}_1 = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 \\ 16 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 17 \end{pmatrix}$$

$$E_1 = \begin{pmatrix} 6 - 5 \\ 17 - 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 11 \end{pmatrix}$$

$$\nabla k_1 = \frac{1}{2} \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} 1 \\ 11 \end{pmatrix} = \begin{pmatrix} 22.5 \\ 56.5 \end{pmatrix}$$

$$\nabla B_1 = \begin{pmatrix} 0.5 \\ 5.5 \end{pmatrix}$$

$$k_1 = k_0 - 0.1 \nabla k_1 = \begin{pmatrix} -3.25 \\ -3.65 \end{pmatrix}$$

$$B_1 = B_0 - 0.1 \nabla B_1 = \begin{pmatrix} 0.95 \\ 0.45 \end{pmatrix}$$

Iteration 2 by Arnold Mugabo

Iteration 2

$$\hat{y}_2 = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -3.25 \\ -3.65 \end{pmatrix} + \begin{pmatrix} 0.95 \\ 0.45 \end{pmatrix} = \begin{pmatrix} -14.2 \\ -49.5 \end{pmatrix} + \begin{pmatrix} 0.95 \\ 0.45 \end{pmatrix} \\ = \begin{pmatrix} -13.25 \\ -49.05 \end{pmatrix}$$

$$E_2 = \begin{pmatrix} -13.25 & -5 \\ -49.05 & -6 \end{pmatrix} = \begin{pmatrix} -18.25 \\ -55.05 \end{pmatrix}$$

$$\nabla W_2 = \frac{1}{2} \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} -18.25 \\ -55.05 \end{pmatrix} = \begin{pmatrix} -120.225 \\ -302.625 \end{pmatrix},$$

$$\nabla B_2 = \begin{pmatrix} -9.125 \\ -27.525 \end{pmatrix}$$

$$W_2 = W_1 - 0.1 \nabla W_2 = \begin{pmatrix} 8.77 \\ 26.61 \end{pmatrix}$$

$$B_2 = B_1 - 0.1 \nabla B_2 = \begin{pmatrix} 1.86 \\ 3.20 \end{pmatrix}$$

Iteration 3 by Thomas Kweya Odongo

Date . . .

M	T	W	T	F	S	S
---	---	---	---	---	---	---

Iteration 3

$$\vec{f}_3 = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} 8.77 \\ 26.61 \end{pmatrix} + \begin{pmatrix} 1-86 \\ 3-20 \end{pmatrix} = \begin{pmatrix} 88.6 \\ 61.1 \end{pmatrix} \text{ Approx.}$$

$$E_3 = \begin{pmatrix} 83.6 \\ 55.1 \end{pmatrix}$$

$$\nabla h_3 = \frac{1}{2} \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} 83.6 \\ 55.1 \end{pmatrix} = \begin{pmatrix} 152.0 \\ 400.9 \end{pmatrix}$$

$$\nabla B_3 = \begin{pmatrix} 41.8 \\ 27.55 \end{pmatrix}$$

$$h_3 = h_2 - 0.1 \nabla h_3 = \begin{pmatrix} -6.43 \\ -13.48 \end{pmatrix}$$

$$B_3 = B_2 - 0.1 \nabla B_3 = \begin{pmatrix} -2.23 \\ 0.45 \end{pmatrix}$$